MTH 232(A and B) Dr. James Ashe
Exam 2 March 11, 2013 **Name:**

Justify your answers and show your work. Unless stated otherwise, all problems are worth 5 points.

Evaluate the following definite integral(s).

1. $\int_{-1}^{2} x^2(x^3+1)^4\,dx$

Let $u = x^3+1$ so $dx = \frac{du}{3x^2}$, $u(-1)=0$, and $u(2)=9$.

$$\begin{aligned} &= \int_0^9 \cancel{x^2} u^4 \frac{du}{3\cancel{x^2}} \\ &= \frac{1}{15}u^5|_0^9 = \frac{9^5}{15} - 0 = \frac{9^5}{15} \\ &= \frac{19683}{5} = 3936.6 \end{aligned}$$

2. $\int_{-2}^{1} \frac{t}{\sqrt{4-t^2}}\,dt$

Let $u = 4-t^2$ so $dt = -\frac{du}{2t}$, $u(-2)=0$, and $u(1)=3$.

$$\begin{aligned} &= \int_0^3 \cancel{t} \cdot u^{-1/2} \frac{du}{-2\cancel{t}} \\ &= -\frac{1}{2}\left(\frac{2}{1}\right) u^{1/2}|_0^3 = -\sqrt{3}\ (\approx -1.73205) \end{aligned}$$

A full 200 gasoline tank begins to leaks out of at the rate of $50-2t$, where t is measured in minutes.

3. How much gasoline leaked out in the first 4 minutes?

$$\int_0^4 50-2t\,dt = \left(50t - \frac{2}{2}t^2\right)|_0^4 = 200-16 = 184 \text{ gallons}$$

The function $v(t) = -\sin(t)$, on $0 \le t \le 3/2\pi$, is the velocity in m/sec of a particle moving along the x-axis.

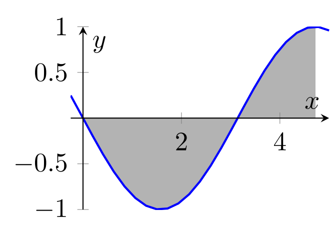
4. [2 points] If the particle has an initial position of $s(0)=1$, find the function $s(t)$ giving its position.

$$\begin{aligned} s(t) &= -\int \sin(t)\,dt \\ &= \cos(t) + C,\ s(0)=1 \Rightarrow C=0 \\ &= \cos(t) \end{aligned}$$

5. Find the total distance traveled over the given interval.

$-\sin(t) = 0$ on $[0, 3/2\pi]$ at $t=0$ and $t=\pi$. So,

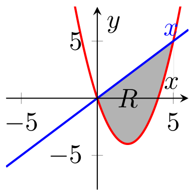
$$\begin{aligned} \text{distance} &= \int_0^{3/2\pi} |-\sin t|\,dt \\ &= \int_0^{\pi} \sin t\,dt + \int_{\pi}^{3/2\pi} -\sin t\,dt \\ &= ((1)-(-1)) + (0-(-1)) = 3 \end{aligned}$$



Use calculus to find the area of the region described.

6. The region bounded by $y = x$ and $y = x^2 - 4x$.

$$\begin{aligned} R &= \int_0^5 x - \left(x^2 - 4x\right)\, dx \\ &= \frac{5}{2}x^2 - \frac{1}{3}x^3|_0^5 = \frac{125}{6} \approx 20.8333 \end{aligned}$$

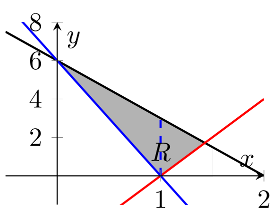


Find the definite integrals(s) which give the area of the region described. ***For this problem you only need to write the simplified integral which gives the area.***

7. The region in the first quadrant bounded by $-3x + 6$, $4x - 4$, and $-6x + 6$. Use calculus

$4x - 4 = -3x + 6 \Rightarrow x = \dfrac{10}{7}$

$$\begin{aligned} R &= \int_0^1 -3x + 6 - (-6x + 6)\, dx + \int_1^{10/7} -3x + 6 - (4x - 4)\, dx \\ &= \int_0^1 3x\, dx + \int_1^{3/2} -7x + 10\, dx \end{aligned}$$



Use the general slicing method and to find the volume of the solid.

8. [8 points] The solid lies between planes perpendicular to the x-axis at $x = 0$ and $x = {}^{\pi}/_{4}$. The cross-sections are squares which are perpendicular to the x-axis and have a height of $\sqrt{\cos 2x}$ measured positively from the x-axis.

The cross-sections are square with sides of $\sqrt{\cos 2x}$. So then the area of each cross-section is $\left(\sqrt{\cos 2x}\right)^2 = \cos 2x$.

$$\begin{aligned} V &= \int_0^{\pi/4} \cos 2x\, dx \\ &= \frac{1}{2}\sin 2x|_0^{\pi/4} = \frac{1}{2}(1) - \frac{1}{2}0 = \frac{1}{2} \text{ cubic units} \end{aligned}$$

Let R be the region bounded by the following curves. Use the disk or washer method to find the volume of the solid generated when R is revolved about the x-axis.

9. $y = \sqrt{x}$, $x = 0$, and $x = 9$.

$$\begin{aligned} V &= \pi \int_0^9 \left(\sqrt{x}\right)^2 \, dx \\ &= \pi \frac{1}{2} x^2 |_0^9 = \frac{\pi 81}{2} \approx 127.235 \text{ cubic units} \end{aligned}$$

For this problem You only need to write the simplified integral which gives the volume.

10. $y = -16x$ and $y = 8x^2 + x^3$.

$-16x = 8x^2 + x^3 \Rightarrow x\left(-x^2 - 8x - 16\right) = x\,(x+4)\,(x+4) = 0$. So $x = 0, -4$

$$\pi \int_{-4}^{0} \left[(-16x)^2 - \left(8x^2 + x^3\right)^2\right] dx = \frac{32768}{35} \approx 3.1019$$

Find the length of the curve.

11. $y = 2x^{3/2}$ from $x = 0$ to $x = \frac{5}{4}$.

$y' = 3x^{1/2}$ and $y'^2 = 9x$.

$\int_0^{5/4} \sqrt{1+9x}\, dx$ let $u = 1 + 9x$. So then $dx = \frac{du}{9}$.

$$\begin{aligned} L &= \frac{1}{9} \int_0^{5/4} u^{1/2} \, du \\ &= \frac{1}{9}\frac{2}{3} u^{3/2} |_0^{5/4} = \frac{2}{27}\left(1 + 9\left(\frac{5}{4}\right)\right)^{3/2} = \frac{335}{108} \end{aligned}$$

BONUS: Use calculus and the general slicing method to find a general formula for the volume of a pyramid with a square base.

Place the pyramid on the Cartesian plane such that the base is perpendicular to the y-axis and the peak touches the x-axis (as we did on the quiz). Let s be length of the base, and h be the height. The equation which gives the distance from the x-axis to the side of the pyramid is the line $y = -\frac{sx}{2h} + \frac{b}{2}$ (the line has slope $\frac{s/2 - 0}{0 - h}$ and y-intercept $(0, b/2)$). So the area of the cross-section is given by

$$A(x) = \left(2\left(-\frac{sx}{2h} + \frac{s}{2}\right)\right)^2 = \left(\frac{s(h-x)}{h}\right)^2 = \frac{s^2}{h^2}\,(h-x)^2$$

and

$$\begin{aligned} V = \int_0^h \frac{s^2}{h^2}(h-x)^2 \, dx &= \frac{s^2}{h^2} \int_0^h (h-x)^2 \, dx \\ &= s^2 \frac{1}{3h^2}\,(h-x)^3 \,|_0^h = s^2 \frac{h^3}{3h^2} = \frac{s^2 h}{3} \end{aligned}$$

The area of the base, $b = s^2$. So this is often written $V = \frac{1}{3} bh$.